

Year Two Lecture Notes

BSc Computer Science - Durham University

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Part I

Artificial Intelligence

1 Module Introduction

Artificial Intelligence has the module code COMP2261, and is led by Dr Yang Long.

1.1 Assessment

This module is assessed through three pieces of summative coursework, these can be seen in Table 1. **Final dates to be confirmed.**

Title	Opening	Due	Feedback	Professor	Weight
Machine Learning	10/10/2026	16/12/2026	22/01/2026	Dr Yang Long	50%
AI Search	16/01/2027	19/02/2027	19/03/2026	Prof. Iain Stewart	25%
Bias in AI	06/02/2027	26/03/2027	07/05/2027	Dr. Shauna Concannon	25%

Table 1: Assessments for Artificial Intelligence.

1.2 Timetable

This module is taught through two one-hour lectures and one two-hour computer class each week as seen below in Table 2. **Final times and locations to be confirmed.**

Day	Time	Class Type	Location	Lecturer
Monday	14:00 - 16:00	Computer Class	TLC025	n/a
Tuesday	16:00 - 17:00	Lecture	W103	Dr Yang Long
Friday	15:00 - 16:00	Lecture	W103	Dr Yang Long

Table 2: Michaelmas Term Timetable for Artificial Intelligence.

2 Machine Learning

3 Artificial Intelligence Search

4 Bias in AI

Part II

Data Science

5 Module Introduction

Data Science has the module code COMP2271 and is led by Dr Frederick Li.

5.1 Assessment

This module is assessed through two pieces of summative coursework and two final exams, these can be seen in Table 3. **Final dates to be confirmed.**

Title	Opening	Due	Feedback	Professor	Weight
Data Collection & Cleaning	31/10/2026	16/01/2027	13/02/2027	Dr Igor Razgon	25%
Image Processing	16/01/2027	26/02/2027	26/03/2027	Dr Amir Atapour...	25%
Probability	May Exam	n/a	n/a	To be confirmed.	25%
Computer Graphics	May Exam	n/a	n/a	To be confirmed.	25%

Table 3: Assessments for Data Science.

5.2 Timetable

This module is taught through two one-hour lectures and one two-hour computer class each week as seen below in Table 4. **Final times and locations to be confirmed.**

Day	Time	Class Type	Location	Lecturer
Monday	12:00 - 13:00	Lecture	W103	Dr Igor Razgon
Friday	10:00 - 11:00	Lecture	W103	Dr Igor Razgon
Friday	16:00 - 18:00	Computer Class	CC-0007	n/a

Table 4: Michaelmas Term Timetable for Data Science.

6 Probability

7 Data Collection and Cleaning

8 Computer Graphics

9 Image Processing

Part III

Networks and Systems

10 Module Introduction

Networks and Systems has the module code COMP2211 and is led by Dr Wanqing Tu.

10.1 Assessment

This module is assessed through two pieces of summative coursework and two final exams, these can be seen in Table 5. **Final dates to be confirmed.**

Title	Opening	Due	Feedback	Professor	Weight
Cyber Security	10/10/2026	06/11/2026	04/12/2026	Dr Charles Morisset	25%
Networks	31/10/2026	08/01/2027	05/02/2027	Dr Wanqing Tu	25%
Distributed Systems	May Exam	n/a	n/a	To be confirmed.	25%
Databases	May Exam	n/a	n/a	To be confirmed.	25%

Table 5: Assessments for Networks and Systems.

10.2 Timetable

This module is taught through one two-hour lecture and one two-hour computer class each week as seen below in Table 6. **Final times and locations to be confirmed.**

Day	Time	Class Type	Location	Lecturer
Thursday	09:00 - 11:00	Lecture	CLC202	Dr Carlton Shepherd
Thursday	16:00 - 18:00	Computer Class	CC-0007	Dr Carlton Shepherd

Table 6: Michaelmas Term Timetable for Networks and Systems.

11 Cyber Security

12 Networks

13 Distributed Systems

14 Databases

Part IV

Programming Paradigms

15 Module Introduction

Programming Paradigms has the module code COMP2221 and is led by Dr Stuart James.

15.1 Assessment

This module is assessed through one piece of summative coursework and two final exams, these can be seen in Table 7. **Final dates to be confirmed.**

Title	Opening	Due	Feedback	Professor	Weight
Systems Programming	31/10/2026	11/12/2026	08/01/2027	Dr Stuart James	50%
Object Oriented Programming	May Exam	n/a	n/a	To be confirmed.	25%
Functional Programming	May Exam	n/a	n/a	To be confirmed.	25%

Table 7: Assessments for Programming Paradigms.

15.2 Timetable

This module is taught through two one-hour lectures and one two-hour computer class each week as seen below in Table 8. **Final times and locations to be confirmed.**

Day	Time	Class Type	Location	Lecturer
Monday	11:00 - 12:00	Lecture	CLC202	Dr Stuart James
Monday	16:00 - 18:00	Computer Class	CB-LG001	n/a
Friday	14:00 - 15:00	Lecture	CLC202	Dr Stuart James

Table 8: Michaelmas Term Timetable for Programming Paradigms.

16 Systems Programming

17 Object Oriented Programming

18 Functional Programming

Part V

Software Engineering

19 Module Introduction

Part VI

Theory of Computation

20 Module Introduction

Part VII

Appendix

21 Artificial Intelligence

21.1 Module Professors

Module Lead: Doctor Yang Long

- Focus: Scalable AI solutions for large-scale healthcare applications.
- Email: yang.long@durham.ac.uk
- Office Hours: **To be confirmed.**

Lecturer: Professor Iain Stewart

- Focus: Computational complexity, graph/group theory, and theoretical aspects artificial intelligence.
- Email: i.a.stewart@durham.ac.uk
- Office Hours: **To be confirmed.**

Lecturer: Doctor Shauna Concannon

- Focus: Computational linguistics, social interaction, and human-computer interaction.
- Email: shauna.j.concannon@durham.ac.uk
- Office Hours: **To be confirmed.**

21.2 References

21.3 List of Figures

22 Data Science

22.1 Module Professors

Module Lead: Doctor Frederick Li

- Focus: Cloud/graphics modelling and remote rendering.
- Email: frederick.li@durham.ac.uk
- Office Hours: **To be confirmed.**

Lecturer: Doctor Igor Razgon

- Focus: Algorithms and complexity, graph theory, and discrete mathematics.
- Email: igor.razgon@durham.ac.uk
- Office Hours: **To be confirmed.**

Lecturer: Doctor Amir Atapour-Abarghouei

- Focus: Machine learning.
- Email: amir.atapour-abarghouei@durham.ac.uk
- Office Hours: **To be confirmed.**

22.2 References

22.3 List of Figures

23 Networks and Systems

23.1 Module Professors

Module Lead: Doctor Wanqing Tu

- Focus: Computer networks and wireless communications.
- Email: wanqing.tu@durham.ac.uk
- Office Hours: **To be confirmed.**

Lecturer: Doctor Charles Morisset

- Focus: Explainable security, access control, and the security and privacy of smart infrastructure.
- Email: charles.morisset@durham.ac.uk
- Office Hours: **To be confirmed.**

Lecturer: Doctor Carlton Shepherd

- Focus: Cyber security.
- Email: carlton.g.shepherd@durham.ac.uk
- Office Hours: **To be confirmed.**

23.2 References

23.3 List of Figures

24 Programming Paradigms

24.1 Module Professors

Module Lead: Doctor Stuart James

- Focus: Artificial intelligence, computer vision, human computer interaction, and digital humanities.
- Email: stuart.a.james@durham.ac.uk
- Office Hours: **To be confirmed.**

24.2 References

24.3 List of Figures

25 Software Engineering

25.1 Module Professors

25.2 References

25.3 List of Figures

26 Theory of Computation

26.1 Module Professors

26.2 References

26.3 List of Figures